Challenges in the data economy: Resistance in the data-driven society

From Analytics to Action

Thank you for filling in the mid-term evaluation!

Suggestions/ideas from you:

- Reading seminar: change to more problem-solving oriented or group work
- Cases: scope, data, analysis, deadlines.
- Weekly quizzes on readings/concepts.
- Consultations with TAs and groups
- And thank you for the positive feedback, too!

Today: Resistance in the data-driven society

- **Learning objectives 4 and 7**

  Compare and assess potential pitfalls and barriers for the project;

  Discuss the project's possible consequences for the market and organisation.

- Understand what “resistance” means in data-driven society
- Recognize how datafication reshapes professional knowledge and judgement
- Identify forms of resistance within technical systems and workplaces

- **Concepts from texts.**
  - Resistance, defensive and productive
  - Resistance tactics
  - Meaningless work

Describe a situation where a digital system forced you to act in a way that felt wrong, inefficient, or meaningless.

Resistance and its goals

- **Resistance** = “intentional actions taken in opposition to someone or something, arising from a refusal to conform to social norms or accept the status quo (or changes to it). It can vary in both the visibility of the act itself and the degree of intent or consciousness behind it (Hollander & Einwohner, 2004).” (Milan, 2024: 2)

- Goal of resistance is “to challenge structural conditions of injustice and inequality, such as laws, state authority, and other forms of power perceived as unjust.” (Milan, 2024: 2)

Resistance and its relevance for data scientists

Where does resistance happen?

- In data collection (what people choose to share)
- In data generation (how they behave because of systems)
- In system interaction (workarounds, etc.)

Why does resistance happen?

- Because data systems a) simplify reality, b) enforce categories, and c) reward certain behaviours over others. → People push back when this clashes with lived reality

Why should you care?

- Resistance degrades data quality
- It produces misleading outputs
- It also signals misalignment between system and reality/values.

Resistance in the data-driven society (Milan, 2024: 3)

Structure of the article:

1. Historical overview of the evolution of resistance.
2. Three key transformations in resistance within a data-driven society: infrastructure, political agency, and tactics.
3. *Six effective resistance tactics for addressing the challenges of datafication*

Main argument:

“This essay argues that the progressive digitalisation and datafication of society, coupled with the rise of intelligent systems (or artificial intelligence, henceforth AI), have fundamentally reshaped the possibilities for resistance.”

Resistance to datafication

- ***Defensive resistance:***
  1. Self-defence
  2. Subversion
  3. Avoidance (impractical as essential services like identity verification require smartphone and app)

- ***Productive resistance:***
  1. Literacy
  2. Counter-imagination
  3. Advocacy campaigning

Overall challenge: Expertise needed to resist harms of data-driven society and ability to confront complexity in matters of concerns.

Data activism (Milan, 2022)

- “**Data activism** represents an emerging form of collective action that invites citizens to take a critical approach toward massive data collection.”

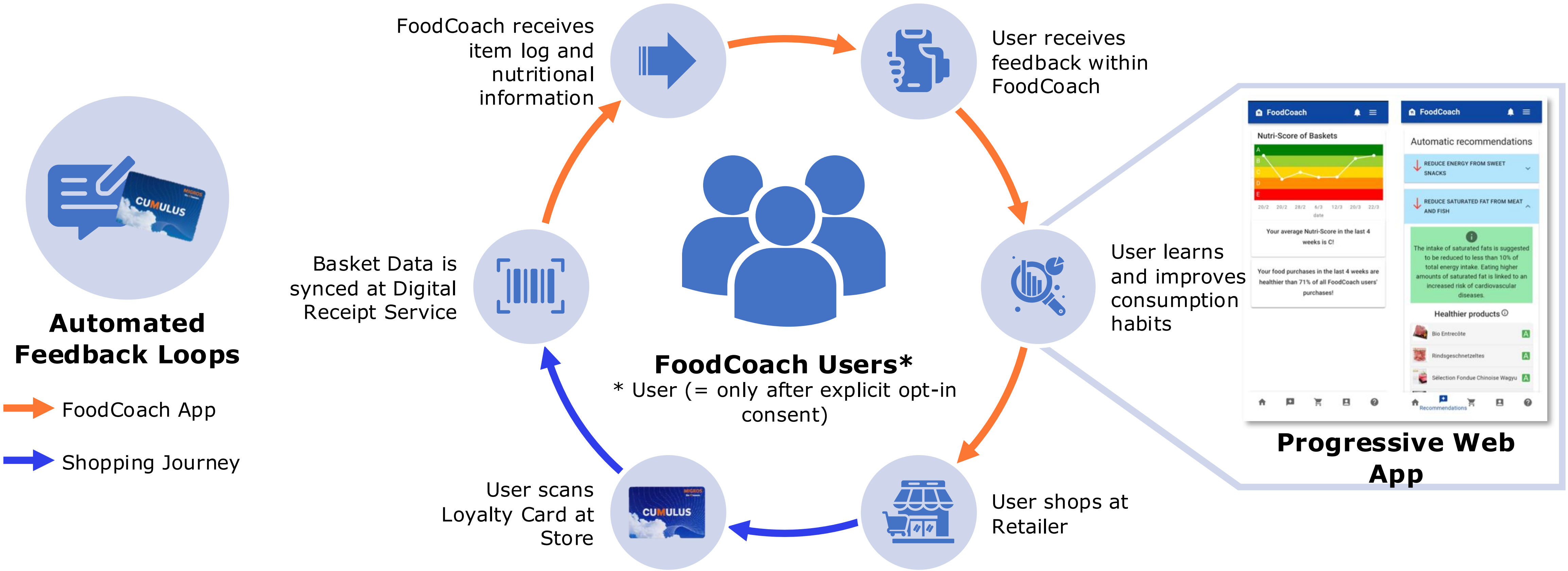
- **Examples of data activists**:
  - Human rights defenders seeking to protect dissidents from government surveillance
  - Environmental activists keeping an independent record of forest depletion
  - Non-governmental organizations meticulously recording cases of police violence against racialized subjects or domestic abuse
  - “Civic hackers” appropriating government data for advocacy purposes

Example 1: FoodCoach research project (2020-2024)

- **Interdisciplinary research project**
- **Digital health**: healthy food shopping and consumption
- **Customer loyalty data**: Automated analysis of digital shopping receipts
- **Progressive web app**: Information is made available to users

Swiss National Science Foundation

FoodCoach Vision: Interventions to change shopping behaviour using digital receipts and food composition databases

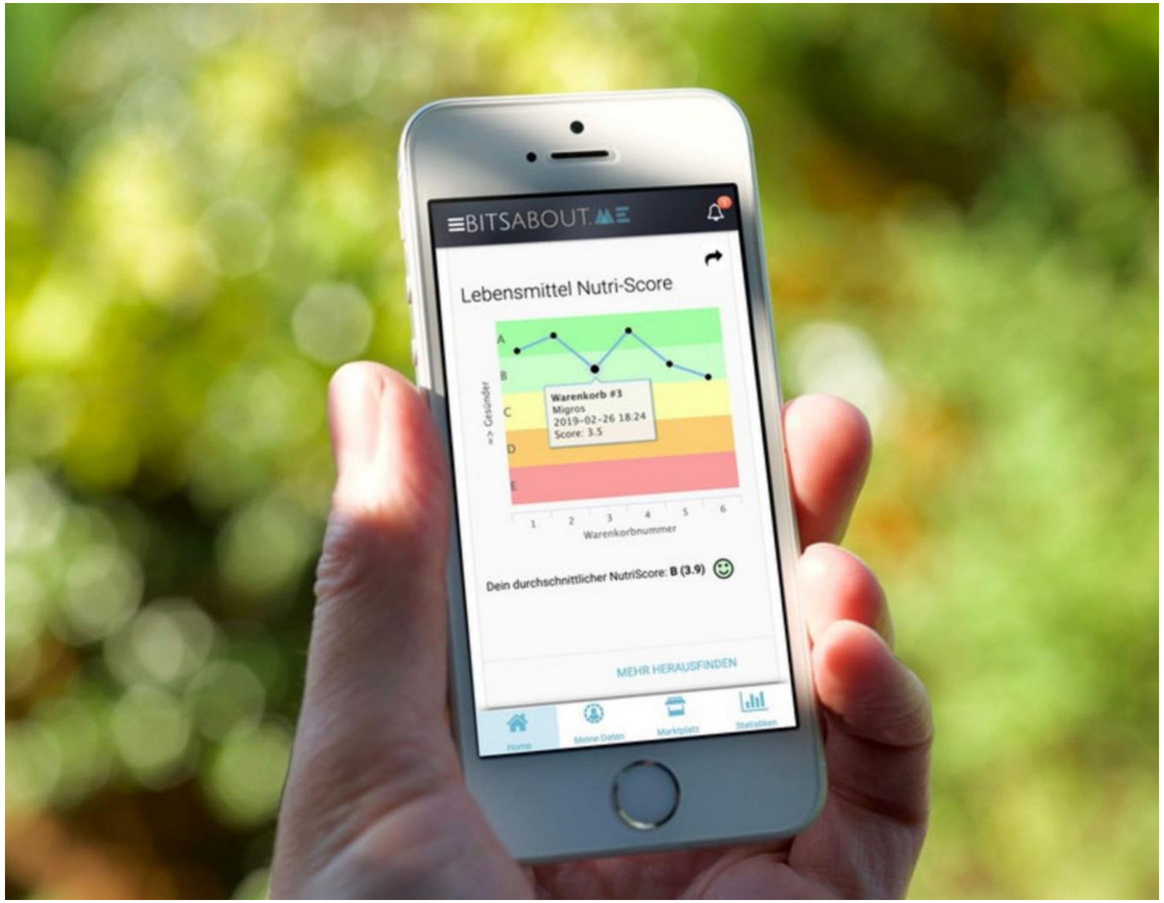


FoodCoach – social science team

Accompanying the **design and development phases** of the FoodCoach app **and integrating user and non-user perspectives.**

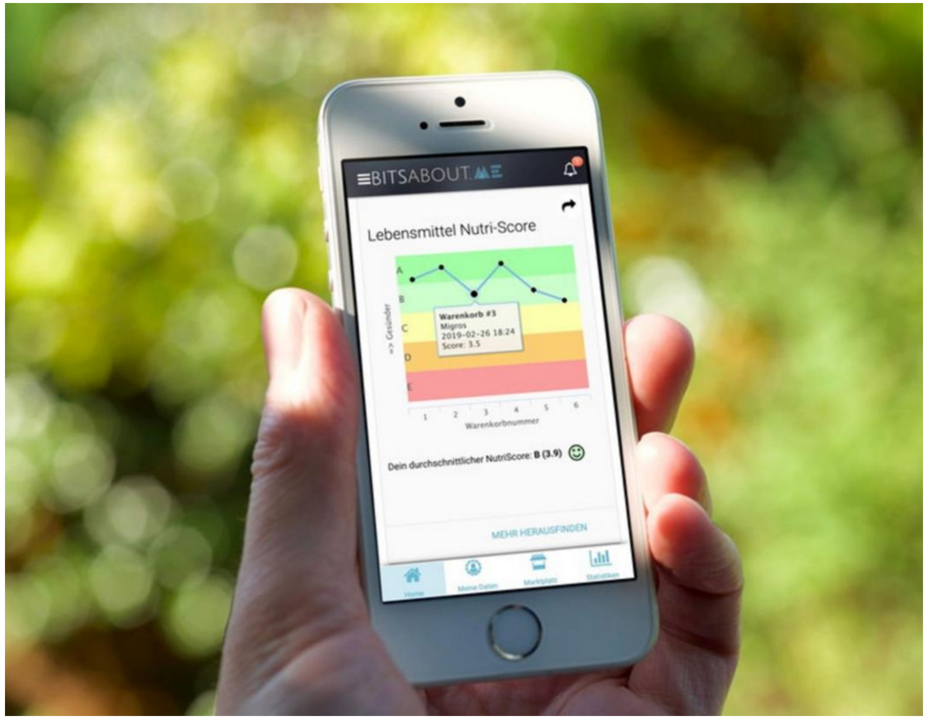
Identify **motivations, intentions, drivers and barriers of potential users and non-users** in relation to digital receipt-based nutrition monitoring

Understanding **new or evolving social norms** of food and digital health.



FoodCoach: Fully Automated Diet Counselling

- ”Unhealthy dietary habits are a major preventable risk factor for widespread non-communicable diseases (NCD). Diet counseling is effective in managing diet-related NCDs, but constrained by its manual nature and limited (clinical) resources. To address these challenges, we propose a fully automated diet counseling system *FoodCoach*. It monitors people's food purchases using digital receipts from loyalty cards and provides structured dietary recommendations. We introduce the FoodCoach system's recommender algorithm and architecture, along with evaluation results from a two-arm randomized controlled trial involving 61 participants. The **trial results demonstrate the technical feasibility and potential for scalable, fully automated diet counseling,** despite not showing a significant change in participants' food purchase healthiness.” (Wu et al., 2025: abstract excerpt)



Lived experiences of digital food tracking

Uses and non-uses of food purchasing data and dietary tracking apps: Living with, beyond, against and despite of them:

- How do people engage or not with mundane food data and technologies (e.g., Kennedy 2018, Lupton 2016, Pink et a. 2017, Hyysalo et al. 2016)

Uses:

- To get informed
- To become aware
- Embodiment though time and sense of empowerment
- Undesirable side effects: excessive self-control, negative feelings (guilty, frustration, disappointment)

Ways of engaging: Playing, cheating and challenging with, addicted to

Non-uses:

- Beyond: abandoning the use after a certain period of time
- Against: persons who ”want not” and “resist” their use
- Despite of: persons who are “socially” or “technically” excluded by their use

Activity (in groups of 3-4): Design, Break, Resistance (1/2)

- You are designing a system to evaluate delivery driver performance using data.
  The company wants a **simple, scalable metric** to optimize productivity.

- Task 1: **Design** a performance system:
  – What data do you collect?
  – What metrics do you use?
  – What counts as "good performance"?

- Task 2: **Break** the performance system (from perspective of delivery driver):
  – How would you game or work around the system?
  – What would you ignore or stop doing?
  – What becomes "meaningless work"?

Activity (in groups of 3-4): Design, Break, Resistance (2/2)

- You are designing a system to evaluate delivery driver performance using data.
  The company wants a **simple, scalable metric** to optimize productivity.

- Task 3: **Discuss** resistance:
  – What is being resisted and why?
  – What does the system fail to capture?
  – What kind of work becomes invisible?

- Task 4: **Debrief** – share in class

- Reflection: If you were redesigning the system, what would you change?

Example 2: MyData

- MyData is a global movement encouraging individual empowerment around personal data.
- Originated in Finland as a response to asymmetries in how data and the means of knowledge production are distributed.
- Aims to shape a more sustainable citizen-centric data economy by means of increasing individuals’ control of their personal data.

Example 3:
Hestia Labs and data collectives (1/2)

“At HestiaLabs, we dream of a digital world where **users take back control over their data**. A digital world where **the benefits generated by data processing are shared with those who produce them**. A digital world where **service providers and users decide together** which data will be used and for what purpose. A digital world where **transparency is the norm**.”

(Source: Hestialabs.org/en/strategy)

<u>Examples:</u> Dating privacy tools for dating app users to understand how their data is used. The Eyeballs reveals invisible trackers that observe, record and analyse everything you do online.

Example:
Hestia Labs and data collectives (2/2)

HestiaLabs
for data collectives

Governing work through personal data: The case of Uber drivers in Geneva (Pidoux, Dehaye, and Gursky)

- Gig workers use data access rights (GDPR) to challenge platform control over their work
- Platform data is hard to interpret: requires technical expertise to make it usable
- Algorithmic management governs pricing, work allocation, and earnings
- Collective data analysis enables worker-led “algorithm auditing”
- Data reveals discrepancies between actual work and paid work: exposing inequalities and supporting legal/political claims (employees)

Link: https://firstmonday.org/ojs/index.php/fm/article/download/13576/11403

What forms of resistance to datafication can you envision related to the cases you work on (Rigshospitalet, Will & Agency)?

‘Meaningless work’ (Hoeyer and Wadmann, 2020)

- The article explores how datafication and digitalizatiaon influence work and goal orientation among professionals in contemporary public organizations (p. 434)

- **Experiences of meaningless work** partly result from technologically mediated ways of making sense of work (p. 435)

- **Organizational environment creating interconnectedness through data**: being seen and recognized as doing important work of sufficient quality implies the use of standardized data that can travel across contexts (p. 444)

Implications of meaningless work

- Financial costs
- Epistemic doubt
- A changed moral landscape: changing work culture and priorities of clinical attention: "'Good data' are no longer justified with reference to their support of clinical outcome" (p. 447)

“Datafication might do most harm when people believe too much in data and forget to retain room for judgement.” p. 449

Synthesis: Where resistance happens

1. Levels of resistance

- Individual (workarounds, ignoring systems)
- Organizational (collective pushback)
- Technical (design choices, system architecture)

2. Targets of resistance

- Data collection
- Data interpretation
- Algorithmic decision-making

3. Motivations

- Protect autonomy
- Preserve meaning
- Contest unfair systems

What could count as meaningless work related to datafication be in the case of Rigshospitalet and Will & Agency?